Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/27/2004
PI Name:	Cheng, Robert K. Ph.D.		
Project Title:	Field Effects of Gravity on Lean Premixed Turbulent Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous flames - premixed		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	06/01/2003	**End Date:**	05/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	C32065J		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

This task is focused on experimental reserach to quantiry the effects of buoyancy on the mean flowfield and its impact on local and overall flame/turbulence interaction processes. The approach is to obtain detailed measurement and analysis of the velocity and scalar fields of premixed turbulent flames in microgravity and in normal, partial and enhanced gravity.

The objective of this research is to proof that buoyancy is a system parameter of laboratory and practical premixed turbulent flames. Buoyancy provides an upstream feedback mechanis that dictates the overall flame flowfield as well as locak flame turbulence interaction processes. Consequently the flame structures, stabilization limits, turbulent reaction rates are directly or indirectly coupled with buoyancy.

Task Description:	The subject of our reserach, lean premixed turbulent flames, is an emerging environmental energy technology because these flames can be exploited to reduce the emission of oxides of nitrogen. Therefore, tesults of this research will benefit both combustion science and technology. Combustion engineer will have scaling information to eliminate buoyancy influences on practical systems. Experimentalists will gain new knowledge to identify and interpret buoyancy contributions to laboratory flames. Theoretical research will benefit by having a set of reference data to improve the fidelity of turbulent combustion models and to develop methods to address elliptical problems. Specifically, this research effort will generate the insight to determine significant processes through which gravity affects flame properties such as flame speed and flame structures. The results can be applied to investigate the gravity-influenced limit for premixed turbulent flames. The backward coupling effects of buoyancy (i.e. downstream effects felt upstream) is of particular importance to combustion theory because it relates to the elliptical problem that requires full time dependent treatment of upstream, wall and downstream boundary conditions. Knowledge on the back-coupling buoyancy affected processes is valuable for guiding the development of turbulent combustion models to include these effects. This study will contribute to resolving the inconsistencies that exist between experiments and theories.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Since the Scientific Concept Review in FY02 the focus of our work has been on developing a Phase B plan for Requirements Definition Review (RDR) with our Projects Scientist (PS) and our Engineering Team Leader of NASA Glenn Research Center (GRC). In our Phase B Plan, we identified 12 open items and described the approaches to resolve them. For the scientific aspects of the flight experiments, the main items are concerned with the hardware, software, and control requirements for the laser diagnostic methods that will enable us to measure conditional velocity statistics and spatial statistical structures of the turbulent flames. We identified that a particle image velocimetry (PIV) is best suited to fulfill these requirement. PIV is a proven method for measuring velocities in a relatively large flowfield domain of 10 cm by 10 cm. From the PIV images, the wrinkled turbulent flame front boundaries are discernible. The flame boundaries can be used as a discriminator for the investigation of conditional statistics. We have developed and tested the software to trace the flame boundaries automatically and to separate and generate the conditional PIV image set. The analysis of the conditioned velocities from the split PIV images, however, is not trivial and requires some software development. We are developing and testing new algorithms for incorporation into the PIV data analysis software. For measuring scalar statistics, our plan is to use a technique called planar imaging of flame fronts (PIFF) based on Mie scattering from an oil aerosol that burns and evaporates at the flame. The main attribute of this PIFF method is its leniency on laser and optical requirements so that it can be implemented by using our PIV laser, optics and camera. To demonstrate, we used our PIV systems to make PIFF measurements in a laboratory v-flame. The data showed this method to produce very clear and detailed outlines of the flame front boundaries that can be extracted and analyzed by the software we have developed previously for the analysis of the flame front measured by a more elaborate diagnostics such as the Planar Laser Induced Fluorescence of OH radicals. Laboratory study of CLEAN flame also is an important aspect of our work to provide background information necessary to interpret data from the flame experiments. Thus far we have used our PIV system to investigate +1g and -1g flames with initial conditions same as those of our flight matrix. The experiments consist of three laminar and three turbulent CH4/air flames with equivalence ratios = 0.7, 0.8, and 0.9 with the mean flow velocity U at 3 m/s. For the turbulent flames, turbulence produced by the grid is slightly anisotropic with u’ = 0.44 m/s and v’ = 0.3 m/s. We are also collaborating on the microgravity combustion project of Prof. Toshihisa Ueda of Keio University in Yokohama Japan. Professor Ueda’s research is to study the relative effects of buoyancy and solid body rotation on the instability of Bunsen type conical flames. The foundation of his project was our previous work on measuring and correlating pulsation frequency of buoyant conical flames. The objective is to extend the investigation to include the study of centrifugal forces on buoyancy effects. Our collaboration has produced two publications.
Bibliography Type:	Description: (Last Updated: 07/09/2019)
Articles in Peer-reviewed Journals	Gotoda H, Maeda K, Ueda T, Cheng RK. "Periodic motion of a Bunsen flame tip with burner rotation." Combustion and Flame. 2004;134:67-79. https://doi.org/10.1016/S0010-2180(03)00082-8 , Jan-2004